

Certificate of Approval

This is to certify that the project proposal entitled “Pharmacy Management System” submitted by Aaditya Chaudhary, Pratima Dahal and Yadav Pathak a student of Bachelor in Computer Application (BCA), 1st Semester, at Gomendra Multiple College has been approved as a bonafide work carried out in partial fulfillment of the requirements for the degree of Bachelor in Computer Application (BCA) under the affiliation of Purbanchal University.

This proposal has been examined and evaluated and is found to be satisfactory in terms of scope, objectives, and methodology as per the academic standards set by the institution.

We hereby recommend this proposal for further development and implementation as a part of the final project work.

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Acknowledgement

I would like to express my sincere gratitude to all those who supported and guided me in completing this project proposal titled “Pharmacy Management System”.

First and foremost, I would like to thank my respected teacher Jhalnath Chapagain for their valuable guidance, encouragement, and constant support throughout the development of this project idea. Their suggestions and feedback were essential in shaping this proposal.

I am also thankful to my friends and classmates who provided helpful insights and motivation during this process.

Finally, I extend my heartfelt thanks to my family for their continuous support and understanding throughout my academic journey.

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Introduction

In today's rapidly evolving healthcare environment, pharmacies play a vital role in ensuring the availability and proper distribution of medicines to patients. Managing medicine inventories, sales records, and stock levels manually can be time-consuming, prone to errors, particularly when dealing with a large variety of medicine products. As a result, there is a growing need for computerized systems that can streamline pharmacy operations and improve record management.

The Pharmacy Management System is a console-based application developed using the C programming language that aims to simplify and automate the management of pharmacy records. The system enables users to store, retrieve, update, and delete information related to medicines, including medicine ID, name, price, stock quantity, and expiry date. It also assists in tracking inventory levels and managing medicine sales efficiently.

This project focuses on applying fundamental programming concepts such as structures, file handling, modular programming, and user-defined functions. Through the development of this system, students gain practical experience in solving real-world problems while enhancing their understanding of data management and software development using the C language.

The Pharmacy Management System is particularly useful for small pharmacies, medical store that require a lightweight and offline solution for maintaining medicine records. The primary objective of this project is to create a reliable, user-friendly, and efficient system that minimizes manual work, improves inventory management, reduces errors and ensure the accurate maintenance of pharmacy data.

Objectives

The primary objective of this project is to design and implement a simple and efficient Pharmacy Management System using the C programming language. The system is intended to automate the process of managing medicine records and pharmacy operations. The specific objectives of the project are as follows:

1. To develop a console-based application that can store and manage medicine records efficiently.
2. To implement CRUD operations – allowing users to Create, Read, Update, and Delete medicine information.
3. To apply file handling in C for storing medicine data permanently in external files.
4. To maintain accurate records of medicines, including medicine ID, name, price, stock quantity, and expiry date.
5. To ensure data accuracy and integrity by preventing duplicate records and validating user inputs.
6. To implement inventory management features such as stock tracking.
7. To facilitate medicine sales and automatically update stock quantities after every transaction.
8. To practice modular programming by using user-defined functions for improved code organization, readability and maintainability.

Problem Statement

In pharmacies and medical stores, maintaining accurate records of medicines is essential for ensuring efficient inventory management and uninterrupted customer service. Traditionally many small pharmacies track medicine information. Stock levels, and sales transactions. While these methods may be suitable for small-scale operations, they often become inefficient and difficult to manage as the number of medicines and transactions increases.

Manual Record-keeping systems are prone to errors such as incorrect data entry, duplicate records, misplaced information, and inaccurate stock calculations. Tracking the availability of medicines, updating stock quantities and searching for specific medicine details can be time-consuming tasks. In addition, monitoring medicines expiry dates and identifying low-stock items becomes challenging when records are maintained manually.

Furthermore, the absence of proper validation mechanisms can lead to inaccurate inventory data, resulting in stock shortages, overstocking, or difficulties in managing medicine sales. Generating bills and maintaining consistent records of medicine transactions also require significant efforts and increase the possibility of human error.

The need for a computerized system that can manage medicine records accurately and efficiently is therefore evident. A console-based application developed using the C programming language can provide an effective solution by automating essential pharmacy operations. The system can offer functionalities such as adding, viewing, searching, updating, and deleting medicine records, along with inventory tracking and sales management features.

The “Pharmacy Management System” aims to address these challenges by providing a reliable, organized, and user-friendly solution for managing pharmacy data. By automating

routine tasks and maintain records through file handling, the system helps improve accuracy, reduce manual workload, save time, and enhance the overall efficiency of pharmacy operations.

Scope of the project

The Pharmacy Management System is a console-based application developed using the C programming language that aims to simplify the management of medicine records and pharmacy operations in small pharmacies, medical stores, clinics, and healthcare centers. The system is designed to help users perform essential operations such as adding in new medicine records, viewing available medicines, searching for specific medicines, updating existing information, deleting records and managing medicine sales.

This project focuses on implementing a single-user, offline system that stores data persistently using file handling techniques, eliminating the need for external databases or complex software environments. It is particularly useful for small-scale pharmacy operations that require a lightweight and reliable solution for maintaining medicine inventories and sales records efficiently.

Key features included within the scope of this project are:

- **Medicine Record Storage:** The system will store medicine information such as medicine ID, name, price, stock quantity, and expiry date in a structured format using files.
- **CRUD Operations:** Users will be able to Create, Read, Update, and Delete medicine records through a simple menu-driven interface.
- **Search Functionality:** The system will allow users to quickly search for medicine records using medicine IDs or medicine names.
- **Inventory Management:** The system will maintain stock information and automatically update inventory levels when medicines are sold.
- **Sales Management:** Users will be able to record medicine sales and generate simple billing information for customers.

- Data Display: The system will provide organized displays of all available medicines and their details.

While the system will not include advanced features such as a graphical user interface (GUI), online connectivity, multi-user access, barcode scanning, or integration with a centralized database, it serves as a practical learning platform for applying fundamental programming concepts in a real-world scenario.

Methodology/Tools Used

The development of the Pharmacy Management System follows a structured programming approach using the C programming language. The project is designed to be a console-based application, suitable for beginners and ideal for practicing core programming concepts. Below is a breakdown of the methodology and tools used in the project:

1. Programming Language

- The system is developed using C, which is well-suited for building low-level, efficient programs.
- Key concepts used:
 - Structured programming: breaking the program into smaller, manageable functions
 - File handling: to store and retrieve medicine records permanently
 - Control structures: loops, conditionals, switch-case, etc.
 - Arrays and structures: to handle multiple medicine records logically

2. Design Methodology

- Modular Design: The project is divided into logical modules such as:
 - Add medicine record
 - Display all records
 - Search a record
 - Modify a record
 - Delete a record
- This approach increases code reusability and makes debugging easier.

3. Development Tools

- Code Editor / IDE:
 - Code:Blocks, Dev-C++, Neovim or Turbo C can be used to write and compile the C code.
- Compiler:
 - GCC (GNU Compiler Collection) or the built-in compiler provided by the IDE
- Operating System:
 - Compatible with Windows, Linux, or macOS, depending on the compiler used.

4. Data Storage Method

- Text file (.txt) is used for storing medicine records.
- File operations include:
 - fopen(), fclose()
 - fprintf(), fscanf()
 - fread(), fwrite() for binary file operations (optional)
- The file system acts as a simple database for maintaining medicine information even after the application is closed.

5. User Interface

- A menu-driven console interface is implemented to interact with the user.
- Options are presented through numbered menus for ease of use.
- Clear prompts, validation messages, and formatted outputs are used to improve usability.

System Requirements

The Pharmacy Management System is designed to run efficiently on standard computer systems with minimal hardware and software prerequisites. Below is the detailed system requirements categorized into hardware and software components:

1. Hardware Requirements

- **Processor:**

A basic Intel or AMD processor with a minimum speed of 1 GHz or higher. Since the application is console-based and lightweight, it does not require a high-performance processor.

- **RAM(Memory):**

At least 512 MB of RAM is sufficient to run the program smoothly. The project is not memory-intensive.

- **Storage:**

The project requires minimal disk space (a few megabytes) mainly for the text files used to store medicine records.

- **Input/Output-Devices:**

Standard keyboard and monitor for user input and output display.

2. Software Requirements

- **Operating-System:**

The program can run on any modern operating system that supports a C compiler, such as:

- Windows (7, 8, 10, 11)
- Linux distributions (Ubuntu, Fedora, etc.)

- macOS
- **Compiler and Development Environment:**

A C compiler and editor are required to write, compile, and run the code. Popular options include:

- Code::Blocks IDE (includes GCC compiler)
 - Dev-C++
 - Turbo C/C++
 - Command-line GCC compiler on Linux/macOS
- **File System Access:**

The program requires permission to read and write files in the local directory for storing medicine records.

3. Additional Requirements

- **Console or Terminal Access:**

Since this is a console-based application, users must have access to a terminal or command prompt window to interact with the program.

- **User Permissions:**

The user should have adequate permissions to create, modify, and delete files.

Modules of the project

The Pharmacy Management System is divided into several functional modules to organize the code logically and to simplify implementation and maintenance. Each module focuses on a specific functionality within the system:

1. Add Medicine Module

- This module allows the user to enter new medicine details such as:
 - Medicine ID
 - Medicine Name
 - Price
 - Stock Quantity
 - Expire Date
- The entered data is then saved to a file for permanent storage.
- Validation checks are performed to prevent duplicate medicine IDs and invalid data entries.

2. Display Medicine Records Module

- This module retrieves and displays all stored medicine records from the file.
- Information is presented in a structured format for easy viewing.
- It enables users to quickly review the current inventory and medicine details.

3. Search Medicine Module

- Enables users to search for a specific medicine using:
 - Medicine ID
- If the medicine exists, the system displays complete information about the medicine.

4. Sell Medicine Module

- This module manages medicine sales transactions.
- Users can select a medicine and enter the quantity to be sold.
- The system automatically:
 - Calculates the total cost
 - Updates the remaining stock quantity
 - Prevents sales when sufficient stock is not available

5. Update Medicine Record Module

- This module allows the user to update existing medicine information.
- The user can search for a medicine by their ID and modify details such as name, Price, or other attributes.
- Changes are saved back to the file, replacing the old data.

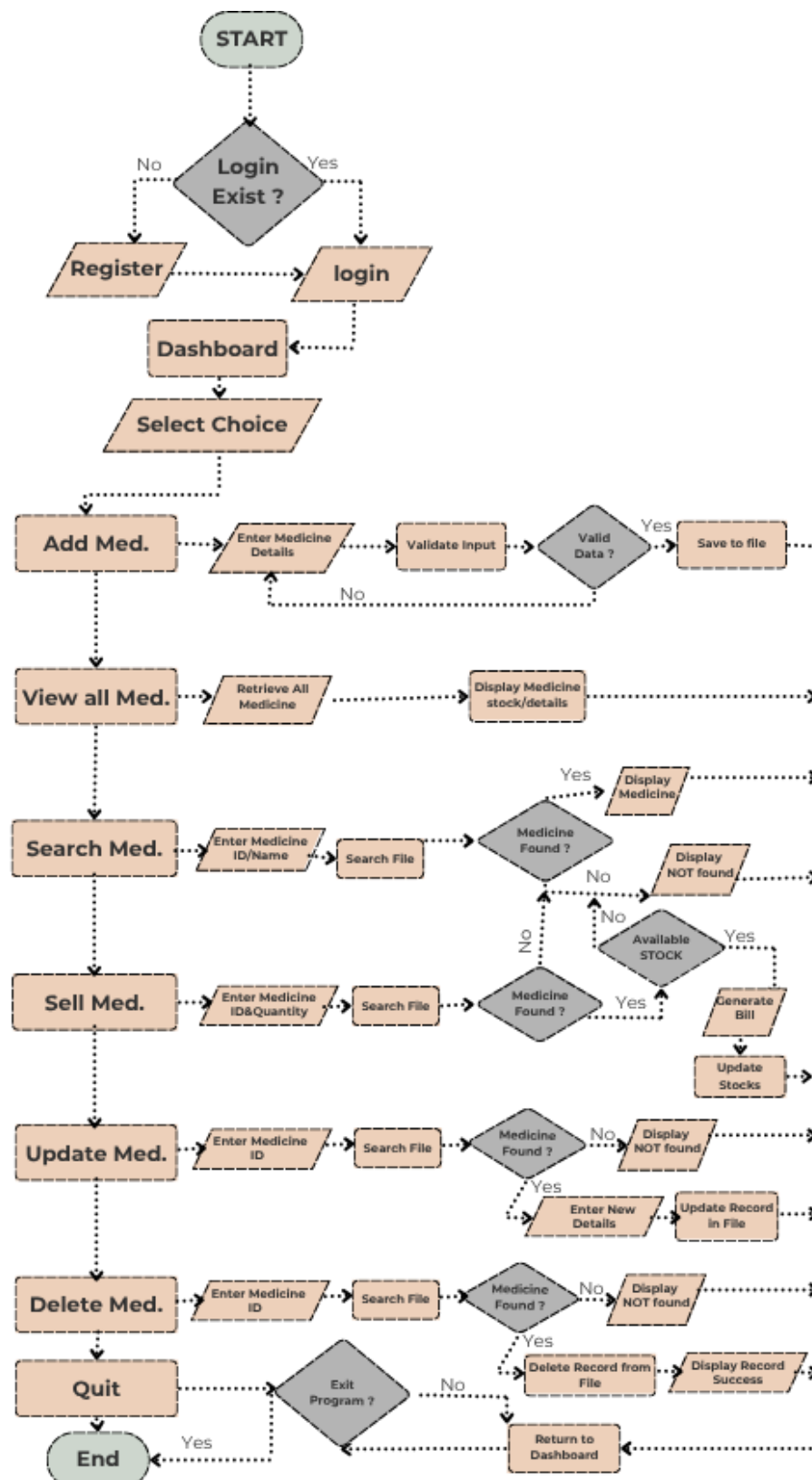
6. Delete Medicine Record Module

- Enables the user to delete a medicine record permanently from the database.
- Medicines can be selected for deletion using their Medicine ID.
- The record is removed from the file, and the inventory is updated accordingly.

7. Exit Module

- Provides a way to safely exit the program.
- Before exit, the program can prompt the user for confirmation to avoid accidental closure.

Flow chart



Algorithm

Step 1: Start

Step 2: Authenticate user login or register a new account.

Step 3: Display dashboard menu options:

1. Add Medicine Record
2. View all Medicine Records
3. Search Medicine Records
4. Sell Medicine Records
5. Update Medicine Records
6. Delete Medicine Records
7. Quit

Step 4: Input user choice.

Step 5: If choice = 1 (Add Medicine record)

- ❖ Input medicine details (ID, Name, price, etc.)
- ❖ Validate Input Data
- ❖ If data is valid, save to file and display confirmation message

Step 6: Else if choice = 2 (View all records)

- ❖ Retrieve all medicine records from file
- ❖ Display medicine stock and details

Step 7: Else if choice = 3 (Search Medicine record)

- ❖ Input medicine ID or Name to search
- ❖ Search the list/database

- ❖ If found, display medicine details
- ❖ Else, display “Not found”

Step 8: Else if choice = 4 (Sell Medicine record)

- ❖ Input medicine ID and Quantity
- ❖ Search file for the record
- ❖ If found, check available stock
- ❖ If stock is available, generate bill and update stocks in file
- ❖ Else, display “Not found”

Step 9: Else if choice = 4 (Update Medicine record)

- ❖ Input medicine ID to update
- ❖ Search file for the medicine record
- ❖ If found, input new details
- ❖ Update the record
- ❖ Display confirmation message
- ❖ Else, display “Not found”

Step 10: Else if choice = 5 (Delete Medicine record)

- ❖ Input medicine ID to delete
- ❖ Search file for the medicine record
- ❖ If found, delete the record
- ❖ Display confirmation message
- ❖ Else, display “Not found”

Step 11: Else if choice = 6 (Exit)

❖ Stop the program

Step 12: Else

❖ Display “Invalid choice”

Step 13: Go back to step 3 (Repeat menu)

Step 14: End

Expected Outcomes

1. Adding Medicine Records

- The system successfully accepts and stores complete medicine information such as Medicine ID, Medicine Name, Price, Stock Quantity, and Expiry Date.
- Validation checks prevent duplicate Medicine IDs and invalid entries.
- Confirmation is displayed after each successful addition of a medicine record.

2. Viewing All Medicine Records

- The system displays a clear and organized list of all stored medicine records.
- Information such as Medicine ID, Name, Price, Stock Quantity, and Expiry Date is presented in a readable format.
- Handles the case when no medicine records exist by displaying an appropriate message.

3. Searching Medicine Records

- The system can accurately search medicine records using Medicine ID.
- When a matching record is found, all relevant medicine details are displayed clearly.
- If no matching record is found, the system informs the user with a “Medicine not found” message.

4. Selling Medicine

- The system allows users to sell medicines by entering the Medicine ID and required quantity.
- The total bill amount is calculated automatically based on the medicine price and quantity sold.

- Stock quantities are updated automatically after a successful sale.
- The system prevents sales when sufficient stock is not available and displays an appropriate warning message.

5. Updating Medicine Records

- The system allows modification of medicine information based on a valid Medicine ID.
- Admin User can update fields such as price, stock quantity, and expiry date.
- A confirmation message is displayed after a successful update.
- If the specified Medicine ID does not exist, the system displays an appropriate error message.

6. Deleting Medicine Records

- The system can remove a medicine record when provided with a valid Medicine ID.
- After successful deletion, a confirmation message is displayed.
- If the Medicine ID is invalid or not found, the system informs the user accordingly.

7. Handling Invalid Inputs

- The system prompts the user when an invalid menu option or incorrect input is entered.
- Validation mechanisms prevent duplicate records, negative prices, invalid stock quantities, and other incorrect inputs.
- Ensures robustness by preventing crashes or data corruption caused by invalid entries.

8. System Exit

- The system terminates cleanly upon user request.

- Before exit, any necessary data saving or cleanup is performed (if applicable).
- All medicine records remain stored in files for future access.

Project Timeline

Phase	Duration
Requirement Analysis	1 Week
Design (Flowcharts/Plan)	1 Week
Coding & Development	2 Weeks
Testing & Debugging	1 Week
Documentation	1 Week
Final Submission	As Scheduled

Conclusion

The Pharmacy Management System is a practical and efficient solution developed to simplify the management of medicine records and pharmacy operations. This project demonstrates how fundamental programming concepts can be applied to address real-world challenges faced by pharmacies and medical stores that rely on manual methods of inventory and record management.

By developing this system using the C programming language, the project highlights the application of essential programming concepts such as structured programming, file handling, conditional statements, looping, and user-defined functions. Through this, it provides the developer with hands-on experience in building a functional application that performs data storage, retrieval, modification, and deletion all crucial for managing records accurately.

The system helps reduce errors, improves accessibility, and saves time compared to traditional manual systems. Users can easily search for records, make updates, or remove outdated data without flipping through piles of documents. It ensures that data is stored permanently using files, making it reusable across multiple sessions without requiring a database system.

Although the project is limited in scope, being a console-based and single-user application—it lays a strong foundation for further development. In the future, the system could be extended to include a graphical user interface (GUI), database integration, multi-user support, and web-based access for broader usability.

Overall, the Pharmacy Management System successfully achieves its intended objectives by providing an effective solution for managing medicine records and inventory in small-scale

pharmacy environments. At the same time, it offers valuable practical experience in software development and strengthens the programmer's understanding of real-world application design using the C programming language.

References

1. **Kanetkar, Yashavant.** *Let Us C* (17th Edition) – BPB Publications
2. This book was an essential reference throughout the development of this project. It provided clear explanations of C programming concepts such as functions, arrays, structures, and file handling, which were directly applied in building this system.
3. **E. Balagurusamy.** *Programming in ANSI C*
4. Another highly useful resource for understanding structured programming techniques, modular design, and standard coding practices in C.
5. **GeeksforGeeks** – www.geeksforgeeks.org
6. A comprehensive programming tutorial website that was used to clarify specific topics such as file operations in C, sorting algorithms, and string manipulation.
7. **TutorialsPoint** – www.tutorialspoint.com
8. Used to review concepts like functions, control structures, and data structures in C. It also helped in testing small code snippets during the development process.
9. **Class Notes and Lecturer Guidance**
10. **Notes** provided during the BCA course and discussions with subject teachers played a key role in understanding the requirements and planning the structure of the system.
11. **YouTube Programming Channels** (e.g., CodeWithHarry, ProgrammingKnowledge)
12. These channels provided visual explanations and sample code implementations, especially helpful during debugging and understanding real-time behavior of the application.
13. **Official C Programming Documentation** – <https://en.cppreference.com/w/c>